

Proposed research timeline

Times		Hours per Week	Goals
Phase 1: Literature methods' replication	Week 1 (June 8th - June 14th)	30	- Refine research question after feedback - Review related work and sources on emergent bias, exploration, adversarial debiasing and reward modeling. Save them for later use Replicate the baseline emergent-bias experiment and the exploration-bonus intervention using closed-model API calls - Create a GitHub repo and begin a weekly progress log.
	<u>Week 2</u> (June 15th - June 21st)	30	- Learn Adversarial ML/RLHF needed for implementations - Decide how each intervention will be tested and compared under the same setup. - Begin introduction + related work - Implement the first version of adversarial debiasing pipeline on closed models.
Phase 2: adversarial debiasing vs. reward modeling in closed models	<u>Week 3</u> (June 22nd - June 28th)	30	- Finish Adversarial debiasing experiments and compare with direct count/exploration baselines. - Start with the reward-modeling setup. Define inputs, labels, evaluation criteria and failure models. - Start goals and methods sections.
	<u>Week 4</u> (June 29th - July 5th)	30	- Run the reward modeling invitation and compare with baseline and adversarial debiasing under the same task, prompts and metrics (the ones used in Wu et al.) - Finish closed model results - Complete introduction, related work, goals, settings and a feasible methods draft..
Phase 3: opening to open weight models	<u>Week 5</u> (July 6th - July 12th)	30	- Research and select open-weight models to test (upon cost, accessibility and relevance to close model results) - Create and set up open model interference pipeline/workflow. - Adapt prompts and experimental code for matching both open and closed models span of results
	<u>Week 6</u> (July 13th - July 19)	30	- Run initial open model experiment - Compare whether emergent bias and mitigation patterns transfer from closed models. - Extend methods section and write results (roughly)
Phase 4: exploration of underlying mechanisms	<u>Week 7</u> (July 20th - July 26th)	30	- Begin discussion: comparisons between interventions - Explore mechanisms using signals like reasoning traces, adversary accuracy and representation methods if possible. - Identify potential useful papers for this matter
	<u>Week 8</u> (July 27th - August 2nd)	30	- Finish underlying mechanism exploration and connect findings to broader fairness question - Complete discussion - Write up and finish conclusion - Clean up the final paper format, add references (and github repo) and add figures (if necessary)